

- STAT 20 SUMMER 2025

3. There is a data set built into R called `mtcars` that includes several measures on different types of cars. Learn more about the data set using `?mtcars`. What is the better way to compare the strength of the linear relationship between these two pairs of variables (`mpg` and `wt`; `mpg` and `hp`): the correlation coefficients or the slopes of the linear models? Why? Explain in 1-2 sentences.
4. Fit the multiple linear regression model from the *Multiple Linear Regression* slides (you may view the previous semester's slides), but instead having "Chinstrap" as the reference level in the species variable. Write the code you used here. Then, interpret the coefficient corresponding to the "Adelie" dummy variable in the context of the problem in 1-2 sentences.